Financial Advisor & Expense Manager AI Agent Development Project – Dual Track Version

Team Size: 3–4 students | **Duration:** 8 Weeks | **Choose Your Challenge Level**

Project Overview

Build an AI-powered financial advisor and expense manager that provides personalized financial advice based on top financial gurus' philosophies while automating expense tracking through payment screenshots and Splitwise data. Users can upload financial books, link articles, and receive tailored advice based on their spending patterns analyzed from payment messages, bank statements, and expense-sharing apps. You'll master financial data processing, OCR technology, and AI-driven financial planning while creating a comprehensive wealth management assistant.

Domain Focus: Financial Technology & Personal Wealth Management
Core Skills: Financial data analysis, OCR processing, expense categorization, financial advisory systems, multi-source financial content synthesis

CHOOSE YOUR TRACK

Track A: Essential (Recommended for Most Teams)

- **Focus:** Core financial advice and expense tracking, achievable goals, strong fundamentals
- **Domain:** Personal Finance Assistant OR Expense Tracking Agent
- **APIs:** 2-3 basic integrations (OCR processing + Financial advice + Basic expense analysis)
- **Framework:** Standard LangChain with financial tools
- **UI:** Streamlit with financial dashboard
- **Deployment:** Single platform (Streamlit Cloud)
- **Features:** Screenshot expense extraction, basic financial advice, spending categorization, simple budgeting

Track B: Advanced (For Confident Teams)

- **Focus:** Enhanced financial planning with intelligent analysis
- **Domain:** Smart Wealth Management Platform
- **APIs:** 3-4 integrations with advanced financial features
- **Framework:** LangChain with custom financial tools
- **UI:** React/Next.js with interactive financial interface
- **Deployment:** Standard cloud deployment with financial data management
- **Features:** Multi-guru financial analysis, automated expense categorization, advanced budgeting, financial goal tracking

Learning Outcomes

By completing this project, you will:

- **Financial Data Processing:** Extract and categorize expenses from various sources (screenshots, messages, files)
- **OCR & Text Extraction:** Process payment screenshots and financial documents automatically
- **Financial Advisory Systems:** Provide personalized advice based on established financial methodologies
- **Expense Analytics:** Create comprehensive spending analysis and budgeting recommendations
- **Multi-Source Content Integration:** Combine financial books, articles, and user data for personalized guidance
- **Financial User Experience:** Design intuitive interfaces for personal finance management

Financial Data & Processing APIs

Primary OCR & Text Processing (Choose 1-2):

- **Google Vision API:** Text extraction from payment screenshots and receipts
- **Amazon Textract:** Advanced document and financial statement processing
- **Tesseract OCR:** Open-source text recognition for payment images
- **Azure Computer Vision:** Multi-language financial document processing

Financial Data Sources:

- **Splitwise API:** Expense sharing and group payment data
- **Bank APIs:** Integration with Indian banks (if available in sandbox)
- **UPI Transaction Data:** PhonePe, Google Pay, Paytm transaction analysis

- **Credit Card APIs:** Statement processing and categorization
- **Manual Upload:** Bank statements, expense receipts, financial documents

Indian Financial Context:

- **Indian Banking:** SBI, HDFC, ICICI bank statement formats
- **UPI Payments:** Integration with popular Indian payment apps
- **Indian Expenses:** Category-specific spending patterns (groceries, transport, dining)
- **Tax Planning:** Indian tax slabs, deductions, and investment options
- **Investment Platforms:** Zerodha, Groww, ET Money integration possibilities

Financial Content & Advice Sources:

- **Financial Books:** PDF processing of popular finance books
- **Guru Content:** Warren Buffett, Robert Kiyosaki, Ramit Sethi principles
- **Indian Finance Experts:** Raghunath Anant Mashelkar, Suze Orman adapted for India
- **Financial Websites:** Content scraping from reliable financial advice platforms
- **Investment Strategies:** SIP, mutual funds, stock market guidance content

Document Processing:

- **PyPDF2:** Financial book and document processing
- **python-docx:** Financial planning templates and reports
- **CSV Processing:** Bank statement and transaction data handling
- **Image Processing:** Receipt and payment screenshot analysis
- **Regular Expressions:** Transaction message parsing and categorization

Week-by-Week Structure

Week 1-2: Foundation & Quick Win

Goal: Get basic expense extraction and financial advice working

Track A Checklist:

- [ ] Create GitHub repo with financial advisor project structure
- [ ] Set up development environment (Python, LangChain, Streamlit, OCR libraries)
- [ ] Build basic image upload and text extraction from payment screenshots

- [ ] Implement simple expense categorization (food, transport, entertainment, etc.)
- [ ] Create basic financial advice database from popular financial principles
- [ ] Build Streamlit interface for screenshot upload and expense tracking
- [ ] Deploy basic version on Streamlit Cloud
- [ ] Record 2-minute demo showing expense extraction from screenshots
- [ ] **Milestone:** Working system that extracts expenses and provides basic financial advice

Track B Checklist:

- [ ] All Track A requirements plus:
- [ ] Set up React/Next.js frontend with financial dashboard
- [ ] Implement advanced OCR with multiple image format support
- [ ] Add financial document processing (bank statements, receipts)
- [ ] Create better expense categorization with machine learning
- [ ] Set up user management for personal financial data

Week 3-4: Core Financial Advisory Architecture

Goal: Transform into intelligent financial advisor with multi-source content integration

Track A Checklist:

- [ ] Integrate 2-3 core tools (expense extraction + financial advice + basic budgeting)
- [ ] Implement financial guru-based advice system (upload financial books/articles)
- [ ] Add Splitwise data integration for group expense analysis
- [ ] Create basic spending pattern analysis and visualization
- [ ] Handle multiple expense sources (screenshots, manual entry, CSV upload)
- [ ] Add simple budgeting recommendations based on spending patterns
- [ ] **Milestone:** System provides personalized financial advice based on spending analysis

Track B Checklist:

- [ ] All Track A requirements plus:
- [ ] Add 3-4 advanced tools (extraction + advice + budgeting + goal tracking)
- [ ] Create intelligent financial content processing from multiple books/sources

- [ ] Implement advanced expense pattern recognition and anomaly detection
- [ ] Add automated categorization with custom category learning
- [ ] Create comprehensive financial health scoring system

Week 5-6: Domain Specialization

Track A Options (Choose ONE):

Option A1: Indian Personal Finance Advisor

- [ ] Add Indian-specific financial advice (PPF, ELSS, SIP recommendations)
- [ ] Implement Indian expense categories and spending patterns
- [ ] Create tax-saving recommendations based on Indian tax laws
- [ ] Add integration with popular Indian payment apps (UPI transaction analysis)
- [ ] Handle Indian financial terminology and currency formatting

Option A2: Automated Expense Manager

- [ ] Create comprehensive expense automation from various sources
- [ ] Add advanced receipt and payment message processing
- [ ] Implement smart expense prediction and budgeting
- [ ] Generate detailed financial reports and spending insights
- [ ] Add goal-based savings and investment tracking

Track B Requirements:

- [ ] Integrate 3-4 financial APIs with sophisticated error handling
- [ ] Add advanced feature: multi-guru financial philosophy comparison OR intelligent investment recommendations
- [ ] Implement comprehensive financial planning with goal tracking
- [ ] Create advanced analytics dashboard with spending insights and trends
- [ ] Add predictive financial modeling and personalized recommendations

Week 7-8: Polish & Production

Track A Checklist:

- [ ] Create professional Streamlit interface with:
  - [ ] Clean expense upload area with multiple format support
  - [ ] Financial dashboard with spending visualization and categorization

    - [ ] Advice interface with guru-based recommendations
    - [ ] Budget tracking with goal setting and progress monitoring
    - [ ] Export functionality (financial reports, budget plans, expense summaries)
- [ ] Add input validation for financial data and expense processing
- [ ] Implement proper error messages for OCR failures and data processing issues
- [ ] Write comprehensive README with setup instructions and API configurations
- [ ] Prepare final demo video (5-7 minutes) showcasing comprehensive financial management
- [ ] **Milestone:** Production-ready personal financial advisor and expense manager

Track B Checklist:

- [ ] Build React/Next.js interface with modern financial UX
- [ ] Deploy with standard cloud hosting (frontend + backend with secure financial data storage)
- [ ] Add comprehensive financial analytics dashboard with advanced insights
- [ ] Implement basic security practices for sensitive financial data
- [ ] Add performance optimization for large financial dataset processing
- [ ] Create detailed user documentation and financial planning guide
- [ ] Prepare comprehensive demo (8-10 minutes) with advanced financial features

Assessment Criteria

Track A Grading (100 points):

- **Financial Processing Functionality (40 points):** Working system with expense extraction + financial advice + basic budgeting
- **Accuracy & Usefulness (20 points):** Accurate expense categorization and relevant financial recommendations
- **User Interface (15 points):** Clean, functional Streamlit app for personal finance management
- **Deployment (15 points):** Stable hosted application with financial data handling
- **Documentation (10 points):** Clear README with financial setup and usage instructions

Track B Grading (120 points + bonuses):

- **Enhanced Financial Architecture (45 points):** 3-4 integrated tools, advanced financial analysis
- **Better Financial UI/UX (25 points):** React interface with engaging financial management features
- **Cloud Deployment (20 points):** Full-stack deployment with secure financial data management
- **Code Quality (15 points):** Clean code, basic testing, financial data optimization
- **Technical Presentation (15 points):** System explanation, financial methodology walkthrough
- **Bonus Features (up to 15 points):** Predictive analytics, investment recommendations, advanced visualizations

Suggested Tech Stacks

Track A Stack:

- **Frontend:** Streamlit with custom components for financial displays
- **Backend:** Python + LangChain + OCR libraries + pandas + matplotlib
- **Database:** SQLite for financial data and user preferences
- **LLM:** OpenAI GPT-3.5-turbo or Google Gemini for financial advice
- **OCR:** Google Vision API or Tesseract for expense extraction
- **Deployment:** Streamlit Cloud
- **Financial APIs:** Splitwise API, basic OCR, financial content processing

Track B Stack:

- **Frontend:** React/Next.js + Tailwind CSS + Chart.js (for financial visualizations)
- **Backend:** FastAPI OR Node.js + LangChain
- **Database:** PostgreSQL OR MongoDB (secure financial data storage)
- **LLM:** GPT-4 or Claude for sophisticated financial analysis and advice
- **OCR:** Advanced OCR with multiple providers and fallback options
- **Deployment:** Vercel (frontend) + Railway/Render (backend)
- **Financial APIs:** Multiple data sources + advanced financial analytics

Success Metrics

Track A Success:

- 90%+ teams deploy working financial advisor with expense tracking
- 80%+ teams demonstrate accurate expense extraction and useful financial advice
- All teams can explain financial data processing and advisory systems
- Portfolio-worthy fintech project for internship applications

Track B Success:

- Enhanced financial platform with intelligent analysis capabilities
- Standard deployment practices with secure financial data handling
- Advanced financial advisory tools and predictive insights implementation
- Interview-ready technical presentation for fintech and financial services roles

Submission Requirements

All Teams Submit:

GitHub Repository with:

- Complete source code with financial workflow automation and data privacy protection
- README with setup instructions, API configuration, and usage examples
- Demo video (5-10 minutes) showcasing comprehensive financial advisor and expense management
- Architecture documentation explaining financial data processing and advisory system flow
- Live deployment with working URL and sample financial scenarios (using mock data)

Viva Presentation covering:

- Financial data processing and OCR implementation strategies
- Technical architecture decisions for handling sensitive financial information
- Financial advisory algorithms and expense categorization techniques
- Individual team member contributions and learning outcomes
- Future improvement ideas for enhanced financial management features

Track B Additional Requirements:

- Technical Blog Post (Medium/LinkedIn)
- Performance Benchmarks for financial data processing and analysis

- Financial data security assessment documentation
- Financial advisory accuracy evaluation report

Portfolio Benefits:

- **LinkedIn Showcase:** Help teams create professional fintech project posts
- **GitHub Stars:** Instructor promotion of best financial advisor repositories
- **Industry Connections:** Introduction to fintech companies and financial services providers
- **Internship Opportunities:** Showcase to companies like Razorpay, Paytm, Zerodha, PolicyBazaar, and fintech startups

Frequently Asked Questions

Q: How do I ensure accuracy in expense extraction from screenshots and messages?
A: Implement multiple OCR providers with fallback options, add manual correction capabilities, use confidence scoring for extraction results, and provide user verification for critical financial data.

Q: What about security and privacy with sensitive financial information?
A: Never store sensitive financial data permanently, use encryption for temporary data, implement proper access controls, provide clear privacy policies, and allow users to delete their data completely.

Q: How do I provide personalized financial advice without being a certified advisor?
A: Base recommendations on established financial principles from recognized experts, include disclaimers about seeking professional advice, focus on educational content, and provide general guidelines rather than specific investment recommendations.

Q: Can we switch tracks mid-project?
A: Yes, from B to A after Week 4 checkpoint. Consider the complexity of advanced financial analytics and multi-source data integration when deciding.

Q: How do I handle different financial philosophies and potentially conflicting advice?
A: Present multiple perspectives clearly, explain the reasoning behind different approaches, allow users to choose their preferred financial philosophy, and provide balanced comparisons of different strategies.

Q: What about integration with real banking and financial APIs?
A: Track A: Focus on manual data upload and CSV processing. Track B: Use sandbox APIs where available, implement secure authentication flows, and handle API limitations and rate limiting properly.

Q: How do I make financial advice relevant for Indian users specifically?
A: Include Indian tax laws, investment options (PPF, ELSS, SIP), banking practices, expense categories relevant to Indian lifestyle, and currency formatting appropriate for Indian context.

Q: What about handling different currencies and international expenses?
A: Track A: Focus on single currency (INR). Track B: Implement basic currency conversion, handle multi-currency expenses, and provide appropriate exchange rate handling for international transactions.

Q: How do I validate the effectiveness of financial advice provided?
A: Track user goal achievement, implement feedback systems, compare recommendations with established financial benchmarks, provide clear reasoning for advice, and maintain transparency in recommendation logic.

Q: What about legal compliance and financial regulations?
A: Include appropriate disclaimers, avoid providing specific investment advice, focus on educational content and general principles, comply with data protection regulations, and encourage users to consult certified financial advisors for major decisions.